

Stock Market Analytics – Analysis Document

1. Objective of the Analysis

The objective of this analysis is to examine historical stock market data to understand price trends, trading volume behavior, returns, and risk characteristics of selected stocks. The analysis supports data-driven evaluation of stock performance and risk-return tradeoffs.

2. Dataset Description

The dataset consists of historical stock market data sourced from Yahoo Finance using the yfinance Python library. The data includes Date, Open, High, Low, Close prices, Trading Volume, and derived metrics such as Daily Return and Cumulative Return.

3. Data Preparation and Feature Engineering

Data was extracted programmatically using Python and cleaned for time-series consistency. Feature engineering was performed to derive Daily Returns, Cumulative Returns, Volatility, and Sharpe Ratio. The processed dataset was then exported for visualization in Power BI.

4. Analysis Performed

Price Trend Analysis was conducted to observe long-term and short-term stock movements. Price and Volume Analysis was used to validate price trends using trading activity. Return Analysis focused on cumulative and average returns, while Risk Analysis measured volatility and risk-adjusted performance using the Sharpe Ratio.

5. Key Insights

Stock prices exhibit varying trends over time, indicating different performance levels. Higher trading volume often confirms strong price movements. Stocks with higher returns tend to show higher volatility. Risk-adjusted performance varies across stocks, highlighting the importance of multi-metric analysis.

6. Business Impact

The analysis enables faster evaluation of stock performance, improved understanding of risk-return tradeoffs, and supports informed investment decisions. Automation reduces manual effort and improves analytical efficiency.

7. Conclusion

This project demonstrates an end-to-end financial analytics workflow using Python and Power BI. It transforms raw stock market data into meaningful insights through automation, analytics, and visualization.